

Concept Drift Detection and Adaptive Learning in High Velocity Social Media Streams

Problem Statement

Social media data is continuously generated, and the topics, vocabulary and opinions in the stream can change over time. Because of these changes, a machine learning model trained on older data may gradually lose its performance. This is known as concept drift. In this project, we plan to study concept drift in a social media stream using sentiment classification as the application. The tweets will be processed as a continuous stream instead of a static dataset, and the model will detect changes and adapt itself when drift occurs. We will reproduce an existing adaptive sentiment classification method as the baseline and then add an additional drift detection mechanism to improve its adaptation.

Base Papers

1. Sentiment Classification over Opinionated Data Streams Through Informed Model Adaptation, Vasileios Iosifidis¹, Annina Oelschlager² and Eirini Ntoutsi

[LINK](#)

This paper focuses on sentiment classification when the incoming data changes over time. The authors propose an ageing based approach where older observations gradually have less influence on the model. They also use changes in the vocabulary of the incoming tweets to identify possible concept drift and adjust the model accordingly. The experiments on TwitterSentiment show that adapting the model to changes in the stream can help maintain its classification performance.

Research Question:

How can a sentiment classification model adapt to changes in an evolving opinionated data stream?

Data Used:

TwitterSentiment dataset containing around 1.6 million positive and negative tweets collected over a period in 2009.

Metrics Used:

Accuracy and classification performance over time, along with the behaviour of the detected changes.

Key Findings:

The paper shows that giving less importance to older data helps the classifier adapt to changing streams. The authors use an ageing mechanism and adjust the ageing rate when changes in the stream are detected.

Model:

Multinomial Naive Bayes with an ageing mechanism and vocabulary based change detection.

Role in the project:

This will be our main baseline. We will implement a simplified version of the proposed adaptive MNB method.

2. Concept Drift Awareness in Twitter Streams , Joana Costa*, Catarina Silva*, Mário Antunes*, Bernardete Ribeiro

[LINK](#)

This paper studies how Twitter classifiers behave when the underlying data distribution changes in different ways. The authors create several types of concept drift, including sudden, gradual, incremental and recurring drift, and compare different learning strategies. The results show that there is no single adaptation strategy that behaves identically for every type of drift, highlighting the importance of considering the nature of the changes occurring in the stream.

Research Question:

How do different learning strategies perform when different types of concept drift occur in Twitter streams?

Data Used:

More than 10,000 English tweets collected using selected hashtags. The tweets were arranged into different temporal patterns to simulate sudden, gradual, incremental and recurring drift.

Metrics Used:

Precision, Recall, F1 score and error rate.

Key Findings:

The paper shows that the performance of a learning method depends on the type of drift present in the stream. Time window, ensemble and incremental learning approaches were compared, with incremental learning performing well overall in their experiments.

Model:

SVM with Bag of Words representation, using time window, ensemble and incremental learning strategies.

Role in the project:

This paper will help us design controlled drift scenarios and evaluate how our system behaves under different types of concept drift.

3. Addressing Event-Driven Concept Drift in Twitter Stream: A Stance Detection Application, ALESSIO BECHINI , ALESSANDRO BONDIELLI , PIETRO DUCANGE , FRANCESCO MARCELLONI , AND ALESSANDRO RENDA

[LINK](#)

This paper looks at concept drift caused by real world events in Twitter streams. The authors study vaccination related discussions and observe that major events can change the nature of the incoming tweets. They compare static classification with different adaptation strategies and show that adapting the model to changes in the stream can improve performance. The paper is particularly relevant to our project because it demonstrates how concept drift can naturally occur in social media due to changing events.

Research Question:

How can a Twitter classification system adapt when real world events cause changes in the incoming stream?

Data Used:

A Twitter stream related to vaccination discussions in Italy, containing hundreds of thousands of tweets collected over several years.

Metrics Used:

Accuracy and Macro F measure.

Key Findings:

The study shows that a model trained on earlier Twitter data can lose performance when new events change the discussion. Adaptive approaches and event based adaptation helped maintain classification performance.

Model:

TF IDF with Linear SVM, along with retraining, ensemble and semantic adaptation approaches.

Role in our project:

This paper provides the real world motivation for studying event driven concept drift in social media streams.

Dataset / Stream Source

We plan to use the **TwitterSentiment140 dataset** [\[link\]](#) used in the first base paper. The tweets will be arranged according to their timestamps and replayed as a continuous stream using Apache Kafka. This means the dataset will not simply be loaded once as a CSV. Tweets will continuously arrive through the Kafka topic and will then be processed by the streaming pipeline.

We will also create controlled drift scenarios based on the approach used by paper2. This will allow us to test sudden, gradual and recurring changes where the actual drift point is known.

Chosen Tools

1. **Apache Kafka:** For generating and handling the continuous tweet stream.
2. **Apache Spark Structured Streaming:** For real time preprocessing and stream processing.
3. **Python / Scikit learn:** For text processing and the baseline sentiment classifier.
4. **River:** For online learning and ADWIN based drift detection.
5. **InfluxDB:** For storing streaming performance and drift metrics.
6. **Grafana:** For visualizing accuracy, drift events, latency and throughput.

Enhancement we will do

We will reproduce the vocabulary based drift detection and adaptive ageing approach from Iosifidis et al. as our baseline. We will then add an ADWIN based prediction error detector so that drift can be identified using both changes in the incoming text and changes in the model's prediction errors.

The enhanced system will be compared with the reproduced baseline using:

- Accuracy
- Precision
- Recall
- F1 score
- Drift detection delay
- False alarms
- Recovery time after drift
- Throughput
- Processing latency
- Memory usage

The main comparison will therefore be between the reproduced adaptive MNB baseline and our enhanced MNB + ADWIN approach.